Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
Enterprise Java

Practical – I

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Class : TYIT	Date/Time of Submission : 22/06/2026

AIM(A): Create a simple calculator application using servlet.

1). HTML

```
<html>
  <head>
    <title>Calculator</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initialscale=1.0"> </head>
  <body>
    <form action="Calculator" method="get" name="frm">
      Enter Number 1 :
      <input name="num1" type="text"/></br>
      Enter Number 2 :
      <input name="num2" type="text"/></br>
      Operator
      <select name="Operator">
        <option value="Addition">+</option>
        <option value="Subtraction">-</option>
        <option value="Multiplication">*</option>
        <option value="Division">/</option>
      </select>
      <input type="reset" value="reset"/>
      <input type ="Submit" value="submit"/>
    </form>
  </body>
```

2). sERVLET

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/Calculator"})
public class Calculator extends HttpServlet {

   protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
```

```
        response.setContentType("text/html;charset=UTF-8");

        PrintWriter out = response.getWriter();

        String num1 = request.getParameter("num1");
        String num2 = request.getParameter("num2");
        String op = request.getParameter("Operator");

        int n1 = Integer.parseInt(num1);
        int n2 = Integer.parseInt(num2);

        if (op.equals("Addition")) {
            out.println("Addition is : " + (n1 + n2));

        } else if (op.equals("Subtraction")) {
            out.println("Subtraction is : " + (n1 - n2));

        } else if (op.equals("Multiplication")) {
            out.println("Multiplication is : " + (n1 * n2));

        } else if (op.equals("Division")) {
            out.println("Division is : " + (n1 / n2));
        }

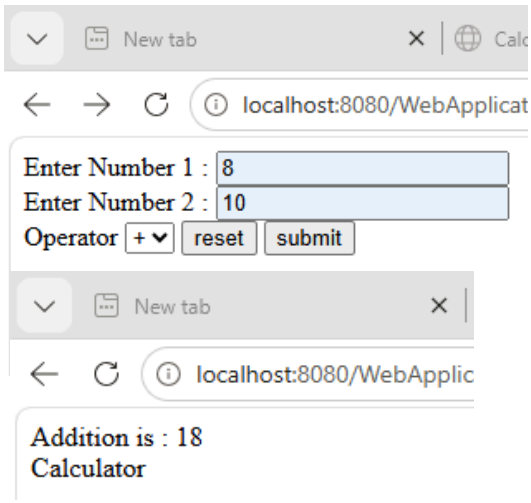
        out.println("<br>Calculator");
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    public String getServletInfo() {
        return "Simple Calculator Servlet";
    }
}
```

OUTPUT:-



(B): Create a servlet for a login page. If the username and password are correct then it says message “Hello <username>” else a message “login Failed”.

1). hTML

```
<!DOCTYPE html>
<html>
<head>
   <title>Login</title>
</head>
<body>

<form action="Practical" method="get">

   Enter username :
   <input type="text" name="uname"><br><br>

   Enter password :
   <input type="password" name="pass"><br><br>

   <input type="submit" value="Submit">
   <input type="reset" value="Reset">

</form>

</body>
</html>
```

2). sERVLET

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
```

```
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/Practical")
public class Practical extends HttpServlet {

    protected void processRequest(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        try (PrintWriter out = response.getWriter()) {

            String uname = request.getParameter("uname");
            String pass = request.getParameter("pass");

            if ("user".equals(uname) && "pwd".equals(pass)) {

                out.println("Hello <b>" + uname
                        + "</b> Welcome to Java Servlet");

            } else {

                out.println("Login Failed");

            }
        }
    }

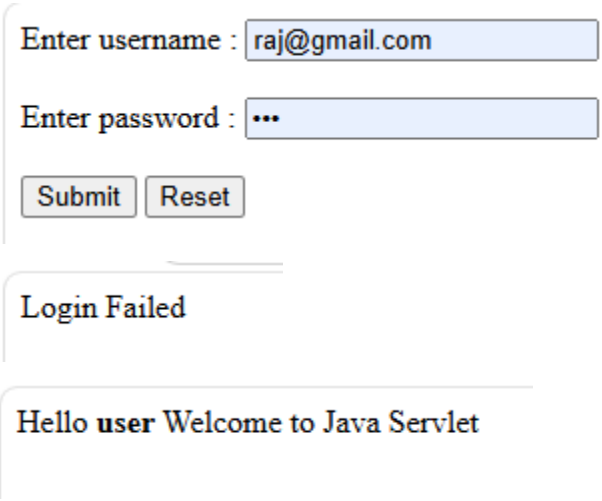
    @Override
    protected void doGet(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {

        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request,
            HttpServletResponse response)
            throws ServletException, IOException {

        processRequest(request, response);
    }
}
```

OUTPUT:-



(C): Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email, and Country from the user using HTML Form and store the registration details in the database.

1.Html code:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Practical 1 C</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <body>
    <form action="prac1c" method="post">
      <table cellpadding="1">
        <thead>
          <tr>

            <th><b>REGISTRATION FORM</b></th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <td><b>First Name : </b></td>
            <td><input type="text" name="fname" value=""></td>
          </tr>
          <tr>
            <td><b>Last Name : </b></td>
            <td><input type="text" name="sname" value=""></td>
          </tr>
```

```
                <tr>
                    <td><b>Zip Code : </b></td>
                    <td><input type="text" name="zip" value=""></td>
                </tr>
                <tr>
                    <td><b>User Name : </b></td>
                    <td><input type="text" name="uid" value=""></td>
                </tr>
                <tr>
                    <td><b>Password : </b></td>
                    <td><input type="password" name="pwd" value=""></td>
                </tr>
                <tr>
                    <td><b>Confirm Password : </b></td>
                    <td><input type="password" name="pwd1" value=""></td>
                </tr>
                <tr>
                    <td><b>Town : </b></td>
                    <td><input type="text" name="town" value=""></td>
                </tr>
                <tr>
                    <td><b>Country : </b></td>
                    <td><input type="text" name="country" value=""></td>
                </tr>
                <tr>
                    <td><input type="submit" value="Submit"></td>
                </tr>
            </tbody>
        </table>
    </form>
  </body>
</html>
```

2.Servlet code:

```
import java.io.IOException;

import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.*;

@WebServlet(urlPatterns = {"/prac1c"})
public class prac1c extends HttpServlet {

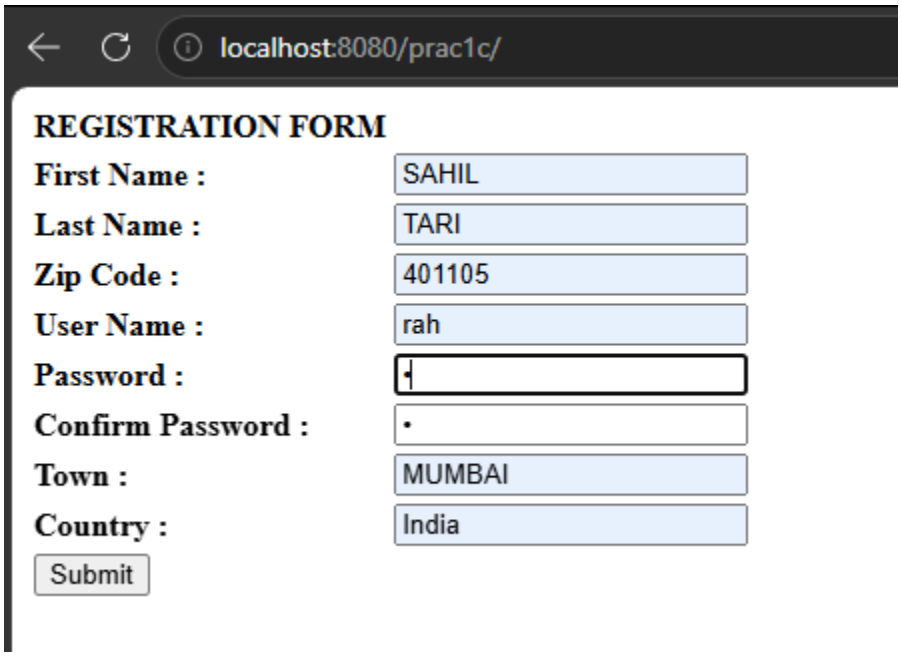
   protected void doPost(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
```

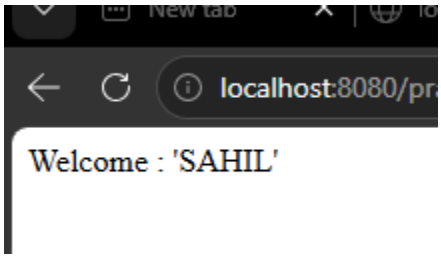
```
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            String connectionURL = "jdbc:mysql://localhost:3306/db";
            Connection connection = null;
            ResultSet rs;
            response.setContentType("text/html;charset=UTF-8");
            String uid = request.getParameter("uid");
            String fname = request.getParameter("fname");
            String sname = request.getParameter("sname");
            String pwd = request.getParameter("pwd");
            String pwd1 = request.getParameter("pwd1");
            String town = request.getParameter("town");
            String country = request.getParameter("country");
            String zip = request.getParameter("zip");
            try {
                Class.forName("org.gjt.mm.mysql.Driver");
                connection = DriverManager.getConnection(connectionURL, "root", "root");
                String sql = "insert into user_register values(?,?,?,?,?,?,?,?)";
                PreparedStatement pst = connection.prepareStatement(sql);
                pst.setString(1, uid);
                pst.setString(2, fname);
                pst.setString(3, sname);
                pst.setString(4, pwd);
                pst.setString(5, pwd1);
                pst.setString(6, town);
                pst.setString(7, country);
                pst.setString(8, zip);
                int numRowsChanged = pst.executeUpdate();
                out.println("Welcome : ");
                out.println("""" + fname + """");
                pst.close();
            } catch (ClassNotFoundException e) {
                out.println("Couldnt Load database driver : " + e.getMessage());
            } catch (SQLException e) {
                out.println("SQL Exception caught : " + e.getMessage());
            } catch (Exception e) {
                out.println(e);
            } finally {
                try {
                    if (connection != null) {
                        connection.close();
                    }
                } catch (SQLException ignored) {
                    out.println(ignored);
                }
            }
        }
    }
}
```

3.Mysql code:

```
CREATE DATABASE db;
USE db;
CREATE TABLE user_register (
uid CHAR(40),
fname CHAR(40),
sname CHAR(40),
pwd CHAR(40),
pwd1 CHAR(40),
town CHAR(40),
country CHAR(40),
zipcode CHAR(40)
);
```

OUTPUT:-





```
mysql> CREATE DATABASE db;
Query OK, 1 row affected (0.00 sec)

mysql> USE db;
Database changed
mysql> CREATE TABLE user_register (
    -> uid CHAR(40),
    -> fname CHAR(40),
    -> sname CHAR(40),
    -> pwd CHAR(40),
    -> pwd1 CHAR(40),
    -> town CHAR(40),
    -> country CHAR(40),
    -> zipcode CHAR(40)
    -> );
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SHOW VARIABLES LIKE 'port';
+---------------+-------+
| Variable_name | Value |
+---------------+-------+
| port          | 3306  |
+---------------+-------+
1 row in set (0.00 sec)

mysql> select * from user_register;
+------+-------+-------+------+------+--------+---------+---------+
| uid  | fname | sname | pwd  | pwd1 | town   | country | zipcode |
+------+-------+-------+------+------+--------+---------+---------+
| rah  | SAHIL | TARI  | sss  | sss  | MUMBAI | India   | 401105  |
+------+-------+-------+------+------+--------+---------+---------+
1 row in set (0.02 sec)
```